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1. INTRODUCTION

1.1. Background and Problem Statement

The use of greenhouses makes it possible to create favorable growing conditions by regulating important factors like temperature, humidity, moisture content in the soil, illumination, and air quality. Such a technique ensures better crop production, protection of plants from unfavorable environmental conditions, and allows production all year round. Yet, to maintain the ideal internal conditions, one must conduct regular control and make timely corrections.

In the past, the process of monitoring the environment in the greenhouse was conducted manually, when farmers checked the condition and made necessary adjustments regularly. It is a tedious and laborious process that often provides no precise information. Failure to detect unfavorable conditions such as dry soil, high temperature, or accumulation of gases may cause problems with growing plants. As more pressure is placed on sustainable and precise agriculture, there is an urgent need for intelligent systems capable of monitoring the environment and providing real-time data. Wireless Sensor Networks (WSNs) appear to be one solution in such scenarios as they help distribute sensor nodes and enable wireless communication among them.

Current monitoring systems are subject to a number of limitations such as:

- High energy consumption levels that make them impractical in battery-powered solutions.
- Dependence on WiFi, which leads to inefficiency when used in remote locations.
- Scalability issues while implementing additional sensor nodes.
- Expensive to implement in small-scale projects.

For this reason, the objective of this project is to design a power-efficient and scalable WSN-based greenhouse monitoring system that implements hybrid communication techniques.

1.2. Objective

In the current project, the main target is to develop a low-power Wireless Sensor Network (WSN) which will be used to monitor the greenhouse conditions in a more effective manner. In addition, the proposed WSN system should work effectively and efficiently with respect to power consumption.

Objectives of this project are as follows:

- Design of wireless sensor nodes for monitoring the greenhouse environment.
- Collection of important parameters such as soil moisture level, ambient temperature, humidity, light intensity, and gas levels.
- Development of an approach for saving power by making use of deep sleep mode.
- Implementation of communication protocol ESP-NOW for communication among sensor nodes.
- Implementation of the wireless network with WiFi technology for establishing remote connection with cloud.
- Designing the system with an easy-to-use GUI supported by an OLED screen.
- Development of an expandable system for the addition of new sensors in the future.
- Developing the WSN system cost-effectively so that it can be used even in small- or medium-sized greenhouses.

1.3. System Overview and Key Component

The suggested design of the WSN includes three major nodes, which include the soil monitoring node, the gas sensing node, and the environmental node that serves as the hub.

The **soil node** has the task of measuring soil moisture using the analog soil moisture sensor. It runs on battery and makes use of deep sleep technology to reduce power consumption. A capacitive touch button will be installed to enable/disable the node.

The **gas sensor node** will be used to collect information about the air quality within the greenhouse. Since gas sensors consume more power compared to others, it also runs on battery and makes use of deep sleep technology to reduce power consumption. Also a capacitive touch button will be installed to enable/disable the node.

The **environmental node** represents the heart of the network. Temperature, humidity (using AHT20) and light intensity (BH1750) sensors will be connected to this node. The environmental node will receive data from the soil node and the gas sensor node using ESP-NOW. Besides collecting the data, the environment node serves as a gateway by linking to WiFi and then sending out the combined data to the cloud via the Blynk platform. Remote monitoring of the system can be done via mobile device. There is also an OLED display incorporated into the environment node which allows for the immediate visualization of the data collected by the sensors.

The **hybrid communication protocol** adopted in the system is:

- ESP-NOW protocol for communication between sensor nodes.
- WiFi protocol for communication with the cloud.

This approach guarantees optimal performance with minimum energy usage and scalability to accommodate future growth in the project.

2. METHODOLOGY

The methodology of the proposed system will involve designing and implementing a WSN with low power consumption for monitoring purposes within a greenhouse environment. The proposed system will have a modular and hierarchical structure that involves a number of distributed sensors (soil, gas) and environment node which serves as the gateway.

2.1. System Design

The proposed system is implemented through a decentralized sensing system where sensor nodes gather information from their surroundings and send them to the main node. This system includes:

- Soil Sensor Node
- Gas Sensor Node
- Environment Sensor (Gateway) Node

Each node acts separately, although they work together logically by transmitting data according to a specific timing system to prevent collision.

The following are the highlights in the implementation process:

- Low energy use
- Real-time data gathering
- Addition of new nodes
- Hybrid communication system (ESP-NOW + WiFi)

2.2. Hardware Implementation

2.2.1. Sensor Nodes (Soil & Gas Nodes)

A typical sensor node is composed of:

- ESP32-C6 Microcontroller (DF-Robot Beetle Mini Dev. Board)

- Soil moisture sensor (Capacitive/Analog type)
or
Gas sensor (MQ Series)
- Battery (18650 3.7V) with charger circuit (TP4056)
- Capacitive touch switch (for ON/OFF)

The nodes are programmed to run in deep-sleep mode and wake up at intervals for data collection and communication.

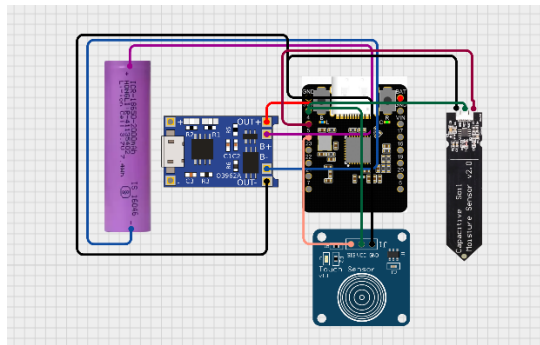


Fig. 1. Soil node circuit diagram

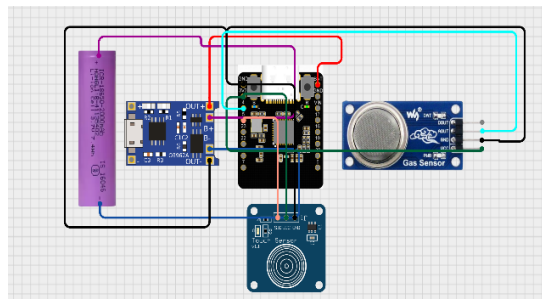


Fig. 2. Gas node circuit diagram

2.2.2. Environment Node (Gateway Node)

The environment node consists of:

- ESP32-C6 microcontroller (DF-Robot Beetle Mini Dev. board)
- AHT20 sensor (temperature and humidity sensor)
- BH1750 sensor (light intensity sensor)
- OLED display for local visualization

This node acts as the central processing and communication hub of the system.

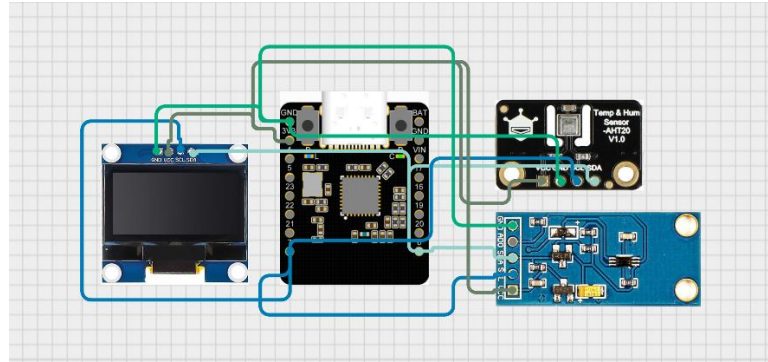


Fig. 3. Environment node circuit diagram

2.3. Communication Methodology

The system uses a hybrid communication model using ESP-NOW + WiFi.

2.3.1. ESP-NOW Communication

Used for communication between sensor nodes and environment node using wireless peer-to-peer protocol. No router or internet is needed.

Each node sends structured data packets containing:

- Node ID
- Sensor readings

To ensure reliable transmission Time-slot based delays are implemented and Acknowledgment handling is managed by ESP-NOW protocol.

2.3.2. WiFi

WiFi is used by the environment node to get internet access and then data is uploaded to the Blynk IoT platform in real time. It allows remote monitoring via mobile application or web dashboard.

This separation ensures that:

- Local communication uses low power (ESP-NOW).
- Cloud communication is solely handled by the gateway node.

2.4. Data Acquisition and Processing

Each sensor node performs the following sequence:

- System wakes from deep sleep.
- Sensor readings are collected.
- Data is transmitted via ESP-NOW to Gateway Node.
- Node returns to deep sleep.

At the environment node:

- Incoming data are received.
- Environmental Data (Humidity, Temperature, Light Intensity) are collected in real time.
- Processed data is displayed onto the OLED and transmitted over the WiFi.

2.5. Data Visualization (Output)

The system provides two levels of visualization, OLED for local level, and Blynk IoT Platform for Cloud Level.

2.5.1. OLED (Local Level)

The OLED display cycles through multiple screens:

- Soil moisture levels.
- Environmental parameters (temperature, humidity, light).
- Gas concentration values.

This ensures real-time local monitoring without external devices.

2.5.2. BLYNK IoT Platform (Cloud Level)

Data is transmitted to the Blynk IoT platform using virtual pins. This allows:

- Remote monitoring via smartphone.

- Real-time dashboard updates.
- Historical data tracking (if enabled)

2.6. Power Management

The sensor nodes are powered via battery and Gateway Nodes are directly powered using USB or bigger Battery. To ensure energy efficiency, the following techniques are implemented:

- Deep sleep mode for sensor nodes. (wake > sense > transmit > sleep)
- Reduced communication overhead using ESP-NOW.
- WiFi usage only at gateway node.

2.7. System Operation Workflow

The complete workflow of the system is as follows:

- Sensor nodes wake at predefined intervals.
- Soil and gas sensors collect data.
- Data is transmitted via ESP-NOW.
- Environment node receives and processes data.
- Local sensors (Humidity, Temperature, Light Intensity) are also read.
- Combined data is displayed on OLED.
- Data is sent to Blynk via WiFi.
- System repeats the cycle.

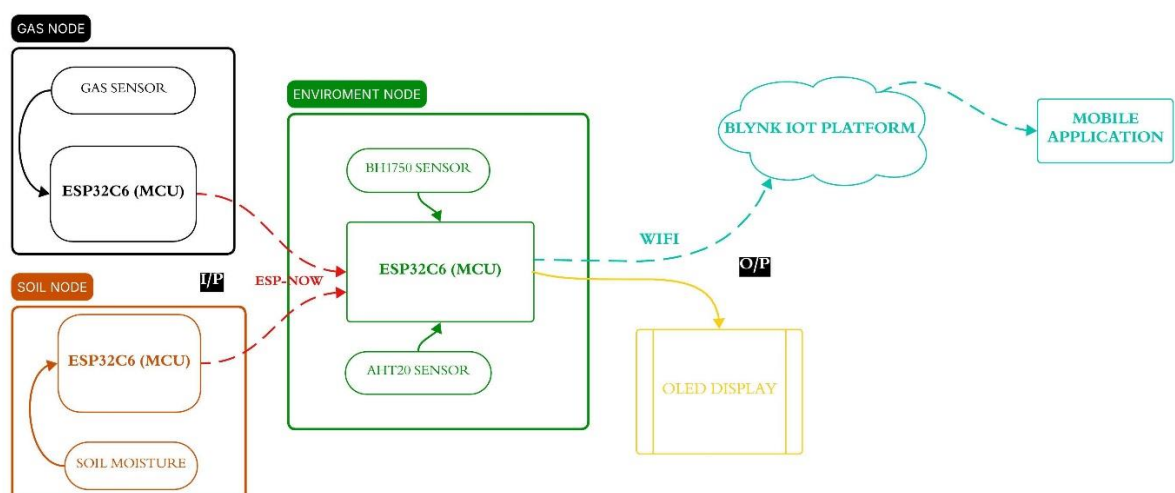


Fig. 4. Block diagram

3. RESULT AND DISCUSSION

The designed model of Wireless Sensor Network (WSN) for greenhouse monitoring achieved success during the real-time testing process. Three different types of nodes were incorporated into the WSN model, including soil moisture sensor node, gas sensor node, and environment sensor node, using an effective hybrid mode of communication, where ESP-NOW is used locally and WiFi connection for cloud.

3.1. Sensor Wise Performance

Soil Moisture Sensor Node:

The soil moisture node detected variations in the moisture content of soil using an analog soil moisture sensor.

- Low moisture content was indicated in dry soil.
- High moisture levels were indicated in wet soil.
- ESP-NOW data transmission worked consistently and reliably.
- Deep sleep mode helped conserve energy.

Thus, the node worked reliably in terms of monitoring soil moisture content and data transmission.

Gas Sensor Node:

This node monitored the quality of air inside the greenhouse using an MQ series gas sensor. The node effectively detected variations in gas concentration under smoke. Some time was needed for warm up to provide reliable readings. The nature of the gas sensor requires it to consume more power than others. Deep sleep mode helps save energy. ESP-NOW data transmission worked consistently and reliably.

Environment (Gateway) Node:

The environment node served as the system center node which collected local parameters such as:

- Temperature using the AHT20 sensor
- Humidity using the AHT20 sensor
- Light intensity using BH1750 sensor

The temperature value was fairly constant and had few variations.

Humidity levels varied appropriately depending on environment changes.

Light intensity values detected low, medium and high illumination levels properly.

Moreover, this node effectively received and processed data from other nodes and displayed data on OLED along with transmitted over WiFi to BLYNK.

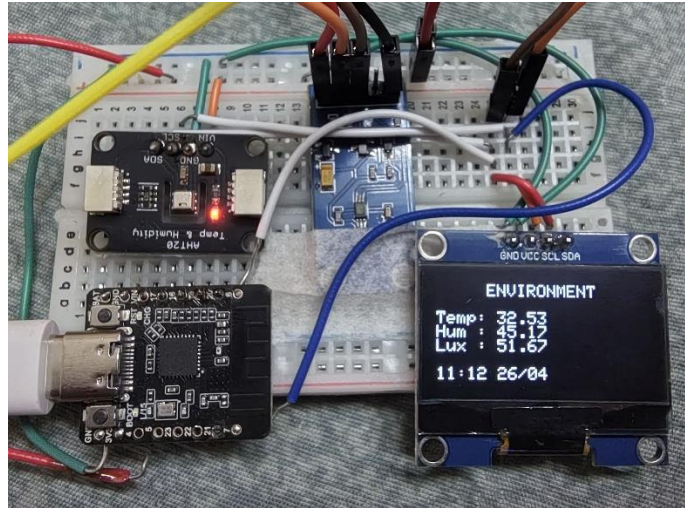


Fig. 5. Environment Node

3.2. Communication Performance

The communication performance of the proposed WSN model was tested under the real-time operating conditions with the hybrid ESP-NOW and WiFi approach.

It was found that the communication between the sensor nodes (soil and gas) and the environment node through **ESP-NOW** was fast and reliable. Data packets were successfully transmitted without a router or internet connection. The time-slot based transmission method helped to avoid packet collision between nodes. In testing there was no significant loss of data observed. The acknowledgement mechanism made sure that data was delivered.

The communication range was enough for small greenhouse environments and the latency was very low, allowing near real-time data transmission. Moreover, ESP-NOW consumes very little power, which was very useful in prolonging the lifetime of the sensor nodes.

The **WiFi communication** was also good in the environment node. The node connected to the network successfully and uploaded sensor data to the cloud platform (BLYNK) in a stable way. There was a small delay in sending the data to the cloud, but it was within the acceptable limits for monitoring applications.

The hybrid communication model was effective overall, integrating low power local communication with reliable cloud connectivity.

3.3. Data Visualization Performance

The system's output devices, including the OLED display and the Blynk IoT platform, were evaluated for accuracy, responsiveness, and usability.

With the **OLED display**, the live data was clearly displayed all the time. It was able to successfully display multiple parameters including the soil moisture, temperature, humidity, light intensity and gas levels. There were smooth transitions between data parameters on the screen, and it was easy to read. The OLED was found to be suitable for local monitoring without any external interface.

The **Blynk IoT platform** was used for remote monitoring which was reliable in tests. The data of the sensors was sent from environment node to the cloud and displayed in dashboard with minimal delay. The mobile interface allowed the users to easily view the real-time values. The graphical visualization of data also made it easier to understand the trends and variations with time.

The two output systems worked together without conflict. Therefore, the OLED provided the local accessibility, and Blynk provided the remote monitoring, thus improving the overall system functionality.



Fig. 6. BLYNK IoT Dashboard (Mobile Application)

4. CONCLUSION, FUTURE SCOPE AND LIMITATIONS

4.1. Conclusion

A hybrid communication model for the Wireless Sensor Network (WSN) based greenhouse monitoring system was successfully designed and implemented. The system was efficient to monitor important environmental parameters such as soil moisture, temperature, humidity, light intensity and gas concentration.

The ESP-NOW protocol enabled low power, fast and reliable communication between sensor nodes, while WiFi enabled seamless cloud connectivity for remote monitoring. Addition of deep sleep mode significantly reduced the power consumption in battery operated nodes, making the system energy efficient.

The environment node served successfully as a center hub for the collection, processing, display and transmission of data. OLED display provided real-time local monitoring, while Blynk platform enabled remote access and data visualization.

The system was found to be overall:

1. Affordable
2. Energy efficient
3. Scalable
4. Reliable for real time monitoring

The project shows a real use case of the IoT and WSN technologies in smart agriculture domain and helps in better management of greenhouse and productivity of the crops.

4.2. Future Scope

The proposed system can be further improved to improve its functionality and real world applicability. Some of the improvements may include:

- 1. Automation of Control System** by adding actuators such as water pumps, and sprinklers for automatic control using sensor data.
- 2. AI-Driven Decision-Making** in smart irrigation control and predictive analysis using ML algorithms.
- 3. Mobile Apps Improvements** by developing of an advanced mobile application with alerts, notifications and user friendly interface.

4. Solar Power Integration by powering sensor nodes using solar panels for more efficient power management.

5. Augmented Sensor Net by adding sensors such as CO2 sensors, pH sensors, and nutrient sensors.

6. Large Scale Deployment by optimizing the system for large agricultural fields and greenhouses.

7. Data logging and analysis by linking the system to the cloud databases for long term storage and comprehensive analysis.

4.3. Limitations

The project has some drawbacks despite its successful implementation:

1. Restricted Range of Communication because ESP-NOW has limited communication range that might not be sufficient for large greenhouses.

2. Dependence on WiFi for Cloud Connectivity which requires WiFi to be always active for remote monitoring of the greenhouse.

3. Battery Usage in Sensor Modules even when optimized, sensor modules running on batteries need to be charged or replaced after a while.

4. Simple (basic)User Interface with an OLED screen and Blynk application.

5. No Control Over Parameters, this system only monitors but doesn't control any environmental or soil parameters.